

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Computer Vision with OpenCV

COURSE CODE: DSA0202

COURSE OUTCOME COVERED

CO5: Apply computer vision techniques to real-world applications such as face recognition, tracking, medical imaging, and autonomous systems. (BL3)

Assessment Tool Weightage: 20%

QUESTIONS: 20

Total Marks:20

Annexure A

Mock Interview

Code of Conduct:

I P Sai Bhavyatha(192424333)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Palacharla Sai Bhavyatha

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge	
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem-solving ability; requires guidance	Unable to solve or apply concepts	
Analytical Thinking	15	Analyzes questions critically and provides well-structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated	
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively	
Confidence	15	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism	
Response to Questions	10	Responds accurately, promptly, and elaborates with relevant examples	Responds correctly but with limited depth	Partial responses; needs prompting	Unable to respond appropriately	

Annexure A

Mock Interview

1. What is Eigenfaces?
2. Define PCA in image processing.
3. What is contour-based shape representation?
4. What is region-based shape representation?
5. How does an active contour (snake) work?
6. How does a level set method work?
7. What is the Hough Transform?
8. How are Fourier descriptors used in shape analysis?
9. How are wavelet descriptors used in shape analysis?
10. What is a particle filter?
11. How is foreground–background separation done in surveillance systems?
12. How do multi-camera systems track pedestrians?
13. What is a Convolutional Neural Network (CNN)?
14. How does a Vision Transformer (ViT) work?
15. What is a Generative Adversarial Network (GAN)?
16. How is a Support Vector Machine (SVM) used for image classification?
17. How are Hidden Markov Models (HMM) applied in activity recognition?
18. What is a Gaussian Mixture Model (GMM)?
19. How is graph-based learning applied to 3D shape correspondence?
20. How is road and lane detection performed in autonomous driving?

ANSWERS

1. What is Eigenfaces?

Answer: Eigenfaces is a face recognition technique based on Principal Component Analysis (PCA). It represents faces using important features called Eigenfaces.

A new face is compared with stored features to identify the person.

2. Define PCA in image processing.

Answer: PCA stands for Principal Component Analysis and is used to reduce image dimensions.

It keeps the most important information while removing less useful features.

It is commonly used in face recognition, compression, and feature extraction.

3. What is contour-based shape representation?

Answer: Contour-based representation describes an object using its outer boundary or outline.

It mainly focuses on the edges and shape of the object.

It is useful for recognizing and comparing objects based on their boundaries.

4. What is region-based shape representation?

Answer: Region-based representation describes an object using its complete internal area.

It uses features such as area, centroid, and moments for shape analysis.

Unlike contour methods, it considers the pixels inside the object as well.

5. How does an active contour work?

Answer: An active contour is a flexible curve placed around an object in an image.

It moves toward the object's boundary using internal and external forces.

It is mainly used for image segmentation and boundary detection.

6. How does a level set method work?

Answer: A level set method represents an object boundary using a mathematical function.

The function evolves gradually so that its zero level represents the changing boundary.

It is useful for image segmentation and handling complex object shapes.

7. What is the Hough Transform?

Answer: Hough Transform is an image processing technique used to detect geometric shapes.

It can identify lines and circles from edge points in an image.

It is commonly used for road marking and lane detection.

8. How are Fourier descriptors used in shape analysis?

Answer: Fourier descriptors represent an object's boundary using frequency components.

They provide a compact mathematical representation of the object's shape.

They are useful for comparing and recognizing different object shapes.

9. How are wavelet descriptors used in shape analysis?

Answer: Wavelet descriptors analyze an object's shape at different scales or resolutions.

They capture both the overall structure and small details of the shape.

They are useful for analyzing objects with complex or irregular boundaries.

10. What is a particle filter?

Answer: A particle filter is a probabilistic method used to track moving objects.

It uses multiple possible positions called particles to estimate the object's actual position.

It is useful when tracking is affected by noise, uncertainty, or occlusion.

11. How is foreground-background separation done?

Answer: Foreground-background separation identifies moving objects by separating them from the background.

Techniques such as background subtraction, frame differencing, and GMM can be used.

It is commonly applied in surveillance systems to detect people and vehicles.

12. How do multi-camera systems track pedestrians?

Answer: Each camera detects and tracks pedestrians within its own view.

Appearance features are extracted and compared between cameras to identify the same person.

This allows continuous pedestrian tracking across multiple camera locations.

13. What is a CNN?

Answer: CNN stands for Convolutional Neural Network and is mainly used for image processing.

It automatically learns features such as edges, textures, shapes, and objects.

CNNs are widely used for image classification, detection, and segmentation.

14. How does a Vision Transformer work?

Answer: A Vision Transformer divides an image into small patches and converts them into feature embeddings.

It uses self-attention to learn relationships between different image patches.

It is commonly used for image classification and other computer vision tasks.

15. What is a GAN?

Answer: GAN stands for Generative Adversarial Network and contains a Generator and Discriminator.

The Generator creates new data while the Discriminator checks whether it is real or generated.

GANs are commonly used for image generation, enhancement, and data augmentation.

16. How is SVM used for image classification?

Answer: SVM is a supervised machine learning algorithm used to classify images into different categories.

It uses extracted image features to find the best separating decision boundary.

It works well when the dataset is small and the features are informative.

17. How are HMMs applied in activity recognition?

Answer: HMM stands for Hidden Markov Model and is used to model activities that occur over time.

It uses observed data to estimate hidden states such as walking, running, or standing.

It is useful for recognizing sequential human activities from video or sensor data.

18. What is a GMM?

Answer: GMM stands for Gaussian Mixture Model and represents data using multiple Gaussian distributions.

Each distribution models a different pattern or component in the data.

It is commonly used for background modeling and foreground detection.

19. How is graph-based learning applied to 3D shape correspondence?

Answer: A 3D shape can be represented as a graph containing nodes and edges.

Graph-based learning compares the relationships between points on different 3D shapes.

It helps identify corresponding points and regions for shape matching.

20. How is road and lane detection performed?

Answer: The camera captures the road and preprocessing is performed to reduce noise and improve the image.

Canny edge detection and Hough Transform can then identify lane markings and road boundaries.

Modern autonomous vehicles also use deep learning for accurate detection in difficult conditions.